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A Systematic Equity Strategy for Mexican Equities and FIBRAs

Primary results for the CNBV-regulated long-only portfolio · long-short and 130/30 variants reported analytically in Section 5.5

Multi-provider data infrastructure, Black–Litterman portfolio construction, machine-learning attribution, and macro-regime conditioning

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Abstract

We document the construction and out-of-sample evaluation of a systematic equity strategy for the joint universe of Mexican publicly traded equities and FIBRAs (the Mexican variant of REITs), with primary results reported for the CNBV-regulated long-only portfolio. The strategy is built on a multi-provider data infrastructure (Bloomberg, Refinitiv, Yahoo Finance), a Black–Litterman portfolio construction layer that combines machine-learning views with low-confidence macro tilts, mean-variance and CVaR optimizers operating under CNBV regulatory constraints, a Layer 2 FX hedge overlay, and a complete suite of overfitting diagnostics including deflated Sharpe ratios and combinatorially symmetric cross-validation. We extend the pipeline with a LightGBM cross-sectional return forecaster, TreeExplainer SHAP attribution, and Banxico macro-regime conditioning. Evaluated on 108 monthly walk-forward out-of-sample rebalances from 2017 through early 2026 with Bloomberg point-in-time fundamentals, the regulated portfolio achieves an annualized Sharpe ratio of 0.03, an annualized return of 8.46% at 13.70% volatility, and a maximum drawdown of −37.19%. ElasticNetCV and LightGBM produce statistically indistinguishable performance on real data; the value of the gradient-boosted model lies in its attribution framework rather than raw return lift. SHAP analysis identifies FIBRA-specific operating metrics — loan-to-value, FFO yield, capitalization rate — as the dominant return predictors, with stability of 0.44 (Spearman) across consecutive rebalances. Regime conditioning reveals that TIGHTENING cycles exhibit higher information-coefficient quality (ICIR 0.59 vs 0.18 in EASING), while EASING cycles produce more stable feature attribution (SHAP stability 0.57 vs 0.41). We argue that the binding constraint on machine-learning approaches to this market is the small effective cross-section of the Mexican universe, and we discuss the implications for live deployment.

Keywords: *quantitative equity, factor investing, FIBRAs, emerging markets, Black–Litterman, gradient boosting, SHAP, macro regimes, walk-forward validation, CNBV regulation.*

1. Introduction

The systematic investment literature has converged on a relatively uniform methodology for developed equity markets: factor portfolios constructed from large cross-sections, ranked by composite scores spanning value, quality, momentum, and low-volatility signals, and reweighted by mean-variance or risk-parity optimization. The methodology works because the cross-section is large enough — typically several thousand stocks in the US and several hundred in major European markets — for the law of large numbers to overwhelm the noise in individual signals. When the same methodology is applied to emerging markets, results typically degrade in proportion to the cross-section size: regional pooling helps, but the per-country signal is fragile.

The Mexican equity market is an extreme case of this problem. The IPC index contains roughly thirty constituents; the float-adjusted investable universe is somewhat smaller. At the same time, the local fixed-income market is deep, the regulatory environment is sophisticated (CNBV operates a position-concentration regime modeled on European UCITS), and the FIBRA market — Mexico's variant of REITs, introduced by reform in 2004 — has matured into a meaningful component of the listed real-asset universe. Mexico is therefore too small to be approached with standard developed-market methods and too institutionally complex to be approached as a pure emerging-market discretionary play. The published systematic literature on the country is sparse.

This paper documents a complete systematic framework calibrated to these specific constraints. The contributions are three. First, the framework integrates a set of FIBRA-specific fundamental features — loan-to-value, FFO yield, capitalization rate, vacancy rate — into a unified cross-sectional model alongside equity fundamentals and price-based signals. These features have no direct analog in standard equity factor libraries, and they are constructed point-in-time from Bloomberg fundamentals with a 90-day reporting lag. Second, the framework layers a Black–Litterman posterior over the cross-sectional model output: per-ticker return views from ElasticNetCV or LightGBM are combined with low-confidence (0.20) macro sector views derived from industrial production, exports, the Banxico overnight target rate, the USDMXN exchange rate, and US inflation. The Black–Litterman step is essential in a small cross-section because it controls the variance of the forecast inputs that feed the optimizer; without it, the mean-variance solution becomes unstable across rebalances. Third, the framework includes a complete suite of overfitting and model-reliability diagnostics: SHAP-based feature stability metrics, Banxico regime conditioning, deflated Sharpe ratios, and combinatorially symmetric cross-validation following Bailey and López de Prado (2014, 2016). The diagnostics are not appendices to the main result; they are the result, in the sense that they tell the operator when the model is trustworthy and when it is not.

The paper proceeds as follows. Section 2 describes the data infrastructure and universe. Section 3 details the feature engineering. Section 4 lays out the modeling methodology. Section 5 presents results. Section 6 discusses what we view as the most important methodological finding of the work — that LightGBM does not materially outperform a well-regularized linear baseline in this universe — and the practical implications for live deployment. Section 7 concludes.

2. Data Infrastructure and Universe

The data layer integrates four providers with automatic fallback. Bloomberg via BLPAPI is the production source, providing point-in-time fundamental data with full historical coverage from 2017 through early 2026. Refinitiv/LSEG is an institutional alternative with somewhat shorter coverage for several constituents. Yahoo Finance is available without credentials but provides only a snapshot of current fundamentals; for the backtest window, only price-based signals (momentum, realized volatility, liquidity) can be constructed from this source. Macro data — Banxico overnight target rate, TIIE 28-day rate, USDMXN, US industrial production, US inflation — is sourced from Banxico SIE and FRED.

The pipeline operates under a strict_data_mode flag set to true in the production configuration. When this flag is enabled, the pipeline aborts on any data failure rather than substituting synthetic or default values. This is the appropriate behavior for a research and pre-production system where silent data substitution is worse than an explicit failure. All fundamental data is lagged by 90 calendar days to approximate the information set available to a real-time investor; restated historical values that would not have been available in real time are excluded.

The investable universe consists of Mexican equities listed on the Bolsa Mexicana de Valores (BMV) and FIBRAs traded on the same exchange. The cross-section comprises approximately twenty-six equities and six FIBRAs at each rebalance date, with modest variation due to corporate actions. The backtest period is monthly rebalances from January 2017 through March 2026 — 108 rebalance dates after warm-up. Returns are total-return, inclusive of dividends and distributions, expressed in Mexican pesos.

Several CNBV regulatory constraints are imposed at every rebalance. Individual positions are capped at 10% of NAV, with a consolidated 10% issuer concentration limit applied across tickers sharing the same issuer ID. A liquidity sleeve of CETES28 and CETES91 instruments occupies 3% to 15% of NAV depending on the prevailing macroeconomic regime, expanding in stress periods (3–5% in expansion, 5–8% in tightening, 8–15% in stress). An optional MBONO3Y buffer of up to 3% is available but disabled in the production configuration. These constraints are fixed structural parameters, not variables in the hyperparameter search.

3. Feature Engineering

Three categories of features feed the cross-sectional model. Equity fundamentals include the price-to-earnings ratio, return on equity, EBITDA growth, profit margin, net debt to EBITDA, capital expenditure to sales, and dividend yield. These are the standard set of value, quality, and capital-allocation factors that the global factor literature has documented (Asness, Frazzini, and Pedersen, 2019; Fama and French, 2015). They are constructed from the most recent reported financials available at the rebalance date with the 90-day lag applied.

FIBRA-specific fundamentals are the central methodological contribution of the feature layer. The FIBRA structure produces a set of operating metrics that have no direct analog in equity reporting and that are routinely used by real-asset specialists to assess relative value. We include four. The capitalization rate is net operating income divided by enterprise value — the FIBRA analog of an earnings yield, but constructed from property-level cash flow rather than accounting earnings. The FFO yield is funds from operations divided by market capitalization, the standard REIT-industry valuation metric that adds back non-cash depreciation to produce a cash-flow yield comparable across capital structures. Loan-to-value is total debt divided by gross asset value, a measure of leverage that for property portfolios has substantially more predictive power for distress risk than the corresponding accounting leverage measures, because property collateral is more directly marked to market than non-real-estate assets. Vacancy rate is the percentage of leasable area not currently under contract — a forward-looking operating metric that has no equivalent in conventional equity reporting.

For equity securities, the four FIBRA-specific features are set to the cross-sectional median (effectively, a neutral exposure). For FIBRAs, equity-specific features that lack operating meaning — return on equity in the manufacturing sense, profit margin on a non-property base — are likewise neutralized. This treatment lets the model identify, empirically, whether the FIBRA features carry pricing information that the equity features cannot replicate; this is the central question of the SHAP analysis in Section 5.2.

Cross-sectional technical features are limited to a 63-trading-day momentum signal and a 63-trading-day realized volatility signal. Short-horizon momentum has well-documented characteristics in emerging-market equities (Cakici, Fabozzi, and Tan, 2013). All features are standardized cross-sectionally within each rebalance date (z-score against the contemporaneous cross-section) before entering the model. Cross-sectional rather than time-series standardization is the appropriate choice for a relative-value framework: the model is asked to forecast which assets will outperform their peers, not whether the overall market will rise or fall.

4. Methodology

4.1 Walk-forward validation

The pipeline is trained and evaluated under strict walk-forward out-of-sample discipline. At each monthly rebalance date t, the cross-sectional model is fitted on data available through t–1, used to generate forecasts for t, and the forecasts are evaluated against realized returns from t to t+1. The training window expands monthly; we do not use a rolling window because the panel is short enough that discarding old observations would meaningfully reduce statistical power.

Hyperparameter selection for the gradient-boosted model uses an inner time-series cross-validation that operates only on the training window, with expanding-window splits and early stopping on the final fold's holdout. The inner CV at rebalance t uses data through t–1 only, with no exposure to information from t or later. Nested validation discipline of this kind is essential: shortcuts such as a single random validation fold or hyperparameter optimization over the full sample are common in the machine-learning-for-finance literature and produce performance estimates that are not realizable in live trading (López de Prado, 2018).

4.2 ElasticNetCV baseline and LightGBM alternative

The baseline cross-sectional model is an elastic-net linear regression with hyperparameter selection by cross-validation. The L1/L2 mixing parameter and regularization strength are selected by the inner time-series CV. The elastic net is the appropriate linear baseline for this problem: it handles the collinearity among fundamental features (which is severe in any fundamental factor set) and the small sample size more gracefully than either ordinary least squares or pure ridge and lasso variants.

The alternative is LightGBM configured as a regressor with mean-squared-error loss. The inner CV scoring metric is the Spearman information coefficient (ic), chosen to align the model-selection criterion with the rank-IC objective of the cross-sectional signal. The model wraps an internal RandomizedSearchCV (5 draws, 3 TimeSeriesSplit folds) over a hyperparameter space spanning tree depth, learning rate, subsample and feature-subsample ratios, L1 and L2 leaf regularization, and minimum child weight. The number of trees is selected dynamically by early stopping on the inner-CV holdout, with a hard cap of 2,000 boosting rounds. The n_jobs = 1 threading configuration is empirically optimal for the small cross-section: OpenMP thread-synchronization overhead (psynch_cvwait) dominates actual compute with ~30 assets, making serial execution 5–11× faster than multi-threaded. Random seeds are fixed across all stochastic components to ensure exact reproducibility.

4.3 Black–Litterman posterior

Forecasts from the cross-sectional model do not feed the portfolio optimizer directly. They are instead converted into per-ticker views and combined with macro sector views in a Black–Litterman posterior (Black and Litterman, 1992). The per-ticker views are assigned confidence weights derived from the model's in-sample fit quality. Macro sector views — derived from industrial production, exports, the Banxico target rate, USDMXN momentum, and US inflation — are blended at low confidence (0.20) specifically chosen so that macro information nudges rather than dominates the quantitative signal. The posterior mean is what enters the optimizer as expected returns.

The Black–Litterman step is essential in a small cross-section. Raw cross-sectional forecasts have substantial variance across rebalances, and feeding them directly into a mean-variance optimizer produces a portfolio that turns over excessively. The BL posterior pulls the expected-return vector toward the equilibrium prior weighted by view confidence, which materially stabilizes the optimizer output without dampening the underlying signal.

4.4 Portfolio optimization

The expected-return vector from the BL posterior enters a portfolio optimizer that operates under CNBV regulatory constraints. Three solvers are available: a sequential least-squares mean-variance optimizer with a market-impact penalty, a min-CVaR optimizer at 95% with a 504-day scenario window, and a Michaud robust solver that averages 100 mean-variance solutions with bootstrap-perturbed expected returns. The covariance matrix is estimated by EWMA with Ledoit-Wolf shrinkage toward the constant-correlation target (Ledoit and Wolf, 2004), using decay $\lambda = 0.94$. Transaction costs are applied at 10 basis points per side on the change in weights at each rebalance.

The production configuration runs the mean-variance and min-CVaR optimizers in parallel for diagnostic comparison; the regulatory NAV reported in this paper uses the mean-variance solution.

4.5 Macro-regime classification

We classify each rebalance into one of three Banxico rate regimes and one of two market stress regimes, combining to up to six cells. The rate regime is TIGHTENING if the Banxico overnight target rate is higher than three months prior, EASING if lower, NEUTRAL otherwise. The stress regime is STRESS if the 60-day realized volatility of the IPC index is in the top quartile of the OOS window, CALM otherwise. Regime labels at time t use only data available at $t-1$. The regime score is smoothed by an EWMA with span 6 and a confidence threshold of 0.30 required for a regime switch.

4.6 SHAP attribution and feature-rank stability

At each rebalance, after the LightGBM model is fitted, we construct a TreeExplainer SHAP instance from the fitted estimator and compute SHAP values for the test slice (Lundberg et al., 2020). These accumulate across rebalances into a panel indexed by (date, ticker, feature, shap_value). The mean absolute SHAP value per feature at each rebalance produces a feature ranking; the Spearman rank correlation of this ranking between consecutive rebalances is our stability metric.

The stability metric serves two purposes. As a global diagnostic, it indicates whether the model's hierarchy of explanatory variables is consistent over time — a stability of 0.80 or higher is typically required for institutional deployment of a tree-based signal. Conditioned on macro regime, it indicates whether the model is more or less reliable in particular environments, which is operationally useful for deciding when to defer to the simpler baseline.

4.7 Hyperparameter optimization

A separate Bayesian search over a broader set of pipeline parameters — Black–Litterman risk aversion, mean-variance and CVaR optimizer parameters, EWMA covariance lambda, and ElasticNet mixing ratios — is run with Optuna's TPE sampler across 25 trials per data source. The objective is a turnover-penalized Sharpe ratio evaluated under purged walk-forward cross-validation with a 21-day gap between training and validation. The hyperopt results are reported alongside deflated-Sharpe and combinatorially symmetric cross-validation diagnostics following Bailey and López de Prado (2014, 2016).

5. Results

5.1 Performance: ElasticNetCV and LightGBM on real data

Table 1 reports the headline out-of-sample performance metrics for both models on each of the three data sources. The figures are for the regulated NAV portfolio under the mean-variance optimizer with CNBV constraints applied; transaction costs are 10 basis points per side, hedge overlay excluded.

Table 1. Out-of-sample performance, regulated NAV, January 2017 – March 2026.

Source	Model	Return	Vol	Sharpe	Sortino	Max DD	CVaR 95%	Turnover
Bloomberg	ElasticNetCV	8.46%	13.70%	0.03	0.03	−37.19%	−1.98%	8.87%
Bloomberg	LightGBM	5.28%	14.56%	−0.17	−0.18	−39.46%	−2.11%	3.49%
Yahoo	ElasticNetCV	3.32%	15.49%	−0.31	−0.31	−43.37%	−2.29%	1.01%
Yahoo	LightGBM	3.39%	15.40%	−0.31	−0.31	−43.11%	−2.27%	1.02%
Refinitiv	ElasticNetCV	2.63%	16.59%	−0.33	−0.33	−48.64%	−2.44%	0.98%
Refinitiv	LightGBM	2.77%	16.28%	−0.33	−0.33	−47.80%	−2.38%	1.33%

Two findings stand out. First, within each data source, ElasticNetCV and LightGBM produce statistically indistinguishable performance. Although the Bloomberg point estimates differ numerically (ElasticNetCV: 0.03 vs LightGBM: –0.17), the 95% confidence interval on the Bloomberg ElasticNetCV Sharpe estimate (paired stationary bootstrap, 5,000 replications) is [–0.634, 0.740] — a band of width 1.37 that dwarfs the numerical gap between the two models. For Yahoo and Refinitiv, where each source's Sharpe rounds to –0.31 and –0.33 respectively, the two models are indistinguishable to two decimal places. There is no statistically meaningful sense in which either model outperforms the other; both results are consistent with zero skill at conventional significance levels on this universe. On Bloomberg, ElasticNetCV carries higher monthly turnover (8.87%) than LightGBM (3.49%), a reversal of the pattern seen with earlier versions of the pipeline.

Second, performance differs substantially across data providers. Bloomberg achieves the highest Sharpe (0.03) because it provides point-in-time fundamental data with full historical coverage; Yahoo Finance and Refinitiv achieve negative Sharpe (–0.31 and –0.33 respectively). For Yahoo Finance, only price-based signals are available because historical point-in-time fundamental data is not accessible through that channel; the absence of fundamental features constrains the signal set. For Refinitiv, coverage gaps in the local-equity feed compress cross-sectional dispersion. The Bloomberg result, while positive, is estimated with very wide uncertainty: the OOS period contains only one full Banxico tightening cycle and one easing cycle, providing limited statistical power.

5.2 SHAP feature attribution

Despite the absence of a performance lift from LightGBM, the SHAP attribution framework produces interpretable and economically sensible results. Table 2 reports the top ten features by time-averaged mean absolute SHAP value across the Bloomberg walk-forward sample.

Table 2. Top features by time-averaged mean |SHAP| value (Bloomberg, LightGBM).

Rank	Feature	Mean |SHAP|	Std |SHAP|
1	ltv	0.00350	0.00363
2	ffo_yield	0.00279	0.00360
3	cap_rate	0.00221	0.00236
4	pe_ratio	0.00196	0.00233
5	dividend_yield	0.00183	0.00214
6	momentum_63	0.00172	0.00249
7	roe	0.00154	0.00183
8	ebitda_growth	0.00146	0.00188
9	profit_margin	0.00145	0.00220
10	capex_to_sales	0.00136	0.00220

Three of the four FIBRA-specific features occupy the top three positions, with loan-to-value alone showing a mean |SHAP| value 1.8 times larger than the highest equity feature (price-to-earnings). The economic interpretation is that FIBRA pricing is driven by operating metrics — property-level cash flow yields, leverage measured against marked-to-market collateral — that have no accounting analog in the equity feature set. A factor model built exclusively on equity-style features would systematically misprice FIBRAs; combining the two universes is informationally productive precisely because the FIBRA features add genuinely orthogonal signal.

The Spearman rank correlation of the feature ranking between consecutive rebalances is reported in Table 3. The mean stability across 107 rebalance pairs is 0.440 for the top five features and 0.428 for the top ten — well below the 0.80 threshold typically required for production deployment. The interpretation is direct: the model's hierarchy of explanatory features reorganizes meaningfully from month to month.

Table 3. SHAP feature-rank stability across consecutive rebalances (Bloomberg, LightGBM).

K	Pairs	Mean Spearman	Std Spearman
Top 5	107	0.440	0.421
Top 10	107	0.428	0.329
All	107	0.455	0.292

We attribute the instability to the small effective cross-section. With approximately thirty assets in the universe, each rebalance's training data provides limited statistical power to pin down feature importance, and the stochastic component of the boosting algorithm amplifies the resulting noise. This is consistent with the finding in Section 5.1 — LightGBM matches but does not exceed ElasticNet on raw performance — and points to the same underlying constraint. Gu, Kelly, and Xiu (2020) document that tree ensembles deliver meaningful performance lift over linear baselines only when the cross-section is large enough that within-period information dominates the noise in feature attribution; the Mexican universe is on the wrong side of that threshold.

5.3 Regime-conditioned performance

Table 4 reports the regime-conditioned metrics, aggregated across stress regimes to focus on the Banxico rate-regime dimension.

Table 4. Performance metrics conditional on Banxico rate regime.

Rate regime	N	IC mean	ICIR	SHAP stab (top-5)	SHAP stab (top-10)
TIGHTENING	64	+0.140	0.59	0.407	0.379
EASING	33	+0.039	0.18	0.566	0.492
NEUTRAL	11	+0.280	0.88	0.386	0.404

Regime metrics computed on walk-forward OOS validation sample; see Table 1 for Bloomberg regulated NAV headline figures. NEUTRAL regime has only 11 observations; treat those statistics as indicative.

The regime picture is more nuanced than a simple easing-vs-tightening ranking. On the signal quality dimension (IC mean and ICIR), TIGHTENING periods are superior to EASING: the mean information coefficient is 0.140 in TIGHTENING versus 0.039 in EASING, and the ICIR (IC mean scaled by IC volatility) is 0.59 versus 0.18. This finding may appear counterintuitive — falling discount rates would be expected to amplify cross-sectional dispersion — but is consistent with the hypothesis that TIGHTENING cycles create more differentiated fundamental trajectories across issuers, particularly between property companies with different loan-to-value profiles facing higher refinancing costs.

On the attribution stability dimension (SHAP), the pattern reverses: EASING cycles produce feature-rank stability of 0.57 versus 0.41 in TIGHTENING. Feature attribution is therefore more reliable in easing environments even though the raw signal quality is lower. The operational implication is that EASING periods offer a better environment for model interpretation and diagnostic monitoring, while TIGHTENING periods offer better raw signal predictability (as measured by IC) with less attribution clarity.

Neither TIGHTENING nor EASING reaches the 0.80 SHAP stability threshold required for institutional deployment. The NEUTRAL regime shows the highest IC mean (0.280) and ICIR (0.88), but with only 11 rebalance observations this result lacks statistical power and should not be extrapolated.

5.4 Hyperparameter optimization and overfitting diagnostics

The Bayesian search over the broader pipeline parameter space (25 trials per source, purged walk-forward cross-validation with 21-day gap) produces best validated Sharpe estimates of 0.15 on Bloomberg, 0.22 on Yahoo, and 0.04 on Refinitiv. These are modest values consistent with the full OOS performance reported in Section 5.1.

The deflated Sharpe ratio adjusts the observed Sharpe for skewness, kurtosis, and the number of configurations tested (Bailey and López de Prado, 2014). Under the null hypothesis of zero skill, the expected maximum Sharpe across 50 trials is 2.23 on the Bloomberg sample (computed via the DSR formula of Bailey & López de Prado, 2014, applied to 50 trials with the skewness and kurtosis of the Bloomberg return series). The observed best validated Sharpe of 0.15 is substantially below this ceiling, which is the appropriate honest finding: the search does not produce evidence of statistically distinguishable skill against the multiple-testing-adjusted null. The PBO statistic via combinatorially symmetric cross-validation is moderate on Bloomberg and low on Yahoo and Refinitiv, indicating that the production parameter choices are robust within the explored space but do not deliver strong evidence of out-of-sample generalization.

These results are not a failure of the search; they are a calibration of expectations. A best-trial Sharpe of 0.15 on a CNBV-regulated, low-turnover, transaction-cost-net portfolio in a small emerging-market universe is a defensible outcome. The deflated-Sharpe adjustment correctly prevents the operator from overstating the result, which is the central function of the diagnostic.

5.5 Layer 2 analytical overlay and reform scenarios

The Bloomberg results with the Layer 2 FX hedge overlay activated are reported on an analytical basis only; they are not included in the CNBV-regulated NAV because the overlay operates with leverage and currency positions that fall outside the CNBV-reportable scope. Bloomberg ElasticNetCV with hedge: Sharpe –0.16, annualized return 3.88%, volatility 24.1%. Bloomberg LightGBM with hedge: Sharpe 0.22, annualized return 13.41%, volatility 22.8%. Neither result reflects the regulated investment outcome; they are reported as a reference for the incremental effect of the overlay on the analytical return stream.

The LFI reform scenario analysis compares the regulated structure against three alternatives: 130/30 long-short, market-neutral, and 130/30 sector-neutral. Scenario-level metrics for each structure are computed in the full pipeline run and are available in reports/output/strategy_report_{source}_{model}.html. The primary regulatory NAV results in Table 1 are the appropriate basis for any compliance-relevant evaluation.

6. Discussion

The most important methodological finding of this work is that the LightGBM cross-sectional forecaster does not materially outperform the elastic-net baseline on real Mexican-market data. This is not a failure of the implementation — both models are constructed with internal walk-forward cross-validation, both are fitted under identical out-of-sample discipline, both are evaluated on the same rebalance sequence — and it is not a surprising result given the constraints of the universe. It is, instead, a useful negative finding that has direct implications for live deployment and for the broader literature on machine learning in emerging-market equities.

The mechanism is the small effective cross-section. Tree ensembles draw their predictive advantage over linear models from the ability to learn non-linear interactions among features without overfitting, but this advantage requires sufficient within-period information for the algorithm to identify those interactions reliably. With roughly thirty assets at each rebalance, the available information per training window is below the threshold at which the algorithm can extract reliable non-linear structure; the model defaults toward the same relationships that the linear baseline already captures, plus added noise from the boosting randomization. The 0.44 SHAP feature-rank stability is a direct measurement of this noise. Gu, Kelly, and Xiu (2020) document this same pattern in cross-country studies: the machine-learning advantage scales with cross-section size, and small universes do not benefit.

This finding does not mean the LightGBM component is without value. The attribution framework — SHAP per rebalance, feature-rank stability, regime-conditioned performance — is genuinely informative independent of whether LightGBM itself generates excess return. The framework tells the operator which features the model is weighting, how those weights move over time, and under what macroeconomic conditions the model is reliable. These are operational diagnostics that the linear baseline cannot produce, and they have value even when the linear baseline matches the gradient-boosted model on raw performance.

Three operational responses follow from these findings. The first is regime-conditional model selection: use the LightGBM forecasts and their attribution during EASING regimes, where SHAP stability reaches 0.57, and rely more heavily on the ElasticNet baseline during TIGHTENING regimes, where raw signal quality is higher (IC mean 0.14, ICIR 0.59) but attribution is less stable. The regime classification is itself a one-period-lagged signal with no lookahead risk. The second is feature pruning: the SHAP decomposition identifies momentum_63 as a contributor to turnover without proportionate contribution to predictive accuracy. Removing or shrinking short-horizon technical features would reduce turnover at modest cost to the information coefficient. The third is ensemble blending: combining ElasticNet and LightGBM predictions at weights inversely proportional to their out-of-sample forecast variance would inherit the stability of the linear baseline and add whatever marginal lift the gradient-boosted model can deliver in favorable regimes.

Several limitations deserve explicit acknowledgment. The strategy is calibrated specifically to the Mexican universe and would not transfer to other emerging markets without recalibration, particularly because the FIBRA-specific features depend on the existence of a deep, liquid REIT segment within the local market. The closest direct analog is the Brazilian Fundos Imobiliários market, where the same methodology could be applied with relatively modest adaptation. A more rigorous treatment of transaction costs would model market impact as a square-root function of trade size relative to daily volume, which would affect the Bloomberg ElasticNetCV configuration disproportionately at its current 8.87% monthly turnover. Finally, the 95% confidence interval on the Sharpe estimate is very wide ([–0.634, 0.740] on Bloomberg) because the nine-year backtest contains only one full Banxico tightening cycle and one easing cycle; precise estimation of regime-conditional performance will require additional years of data or a multi-country extension.

7. Conclusion

We have documented a complete systematic framework for the Mexican equity and FIBRA universe, evaluated under strict walk-forward out-of-sample discipline across three data providers. The framework integrates multi-provider data infrastructure, a Black–Litterman portfolio construction layer that blends machine-learning views with low-confidence macro tilts, multiple optimizers operating under CNBV regulatory constraints, a Layer 2 FX hedge overlay reported on an analytical basis, a LightGBM cross-sectional forecaster with TreeExplainer SHAP attribution, Banxico macro-regime conditioning, and a complete suite of overfitting diagnostics. The regulated portfolio achieves an annualized Sharpe ratio of 0.03 on Bloomberg point-in-time fundamentals over the 2017–2026 window, with a 95% bootstrap confidence interval of [–0.634, 0.740].

The principal empirical findings are three. First, FIBRA-specific operating metrics — loan-to-value, FFO yield, capitalization rate — dominate the SHAP attribution of the gradient-boosted model and carry pricing information that conventional equity factors cannot replicate. Second, LightGBM and ElasticNetCV produce statistically indistinguishable performance on real data within the Mexican cross-section, attributable to the small effective universe; the value of the gradient-boosted model lies in its attribution framework rather than in raw return lift. Third, model reliability is regime-dependent in an asymmetric way: TIGHTENING cycles exhibit higher raw signal quality (ICIR 0.59 vs 0.18), while EASING cycles exhibit higher SHAP feature-rank stability (0.57 vs 0.41) — a distinction with direct operational implications for regime-conditional model governance.

The work points toward three directions for further research. The first is an out-of-region replication of the framework on the Brazilian Fundos Imobiliários market, which has a larger and more diverse REIT cohort and would provide a natural validation of the FIBRA-feature methodology. The second is a more granular macroeconomic conditioning that moves beyond the binary regime classification to explicit term-structure, FX, and commodity factors; the existing regime analysis suggests there is variance to capture that the current binary specification leaves on the table. The third, and most operationally important, is the construction of an ensemble between the ElasticNet and LightGBM paths with regime-conditional weights, exploiting the asymmetric reliability of the two models across the macroeconomic cycle. The framework as constructed provides the diagnostic infrastructure to make such an ensemble robust; the empirical work of fitting it remains to be done.

The strategy in its current form is a research prototype, not a production system. The most useful artifact of this work is not the reported Sharpe ratio but the framework itself: an honest, instrumented, regime-aware approach to a market where naive applications of factor investing and machine learning both fail for diagnosable reasons.

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Appendix A. Hyperparameter Search Space

The LightGBM internal RandomizedSearchCV samples 5 configurations from the following distributions for each training window, with a 3-fold expanding-window TimeSeriesSplit and early stopping (50 rounds) on the inner-CV holdout. The inner-CV scoring metric is the Spearman information coefficient (ic).

Parameter	**Distribution**
max_depth	uniform integer {3, 4, 5, 6}
learning_rate	uniform {0.01, 0.03, 0.05, 0.10}
subsample	uniform {0.7, 0.8, 1.0}
colsample_bytree	uniform {0.7, 0.8, 1.0}
min_child_weight	uniform {1, 5, 10}
reg_alpha	uniform {0, 0.1, 1.0}
reg_lambda	uniform {0.1, 1.0, 10.0}
n_estimators	early-stopped (cap 2,000)

The outer Optuna TPE search operates over a broader pipeline parameter space — Black–Litterman risk aversion (BL τ and risk-aversion), mean-variance and CVaR optimizer parameters, EWMA covariance lambda (bounded [0.90, 0.99]), ElasticNet L1 ratio grid, macro-regime EWMA span, forecast horizon, and ADTV lambda — for 25 trials per data source under purged walk-forward cross-validation with a 21-day gap. The objective is a turnover-penalized Sharpe ratio with penalty weight 0.5.

Appendix B. Software and Reproducibility

All results are reproducible from the repository accompanying this paper. The software stack is Python 3.10 or higher with lightgbm ≥ 4.0, shap ≥ 0.45, scikit-learn, pandas, numpy, Optuna, arch (GARCH), and scipy. Matplotlib and Plotly are used for figure generation; the PDF tearsheet pipeline uses WeasyPrint with an fpdf2 fallback for environments lacking the pango and gobject system libraries. Random seeds are fixed across all stochastic components to ensure exact reproducibility. The test suite comprises 172 unit and integration tests covering walk-forward data integrity, portfolio constraint compliance, SHAP schema, macro-regime no-lookahead, overfitting diagnostics, and bootstrap CI.

Repository: github.com/MaxHidalgoLeon/FondoMexicoAlfa